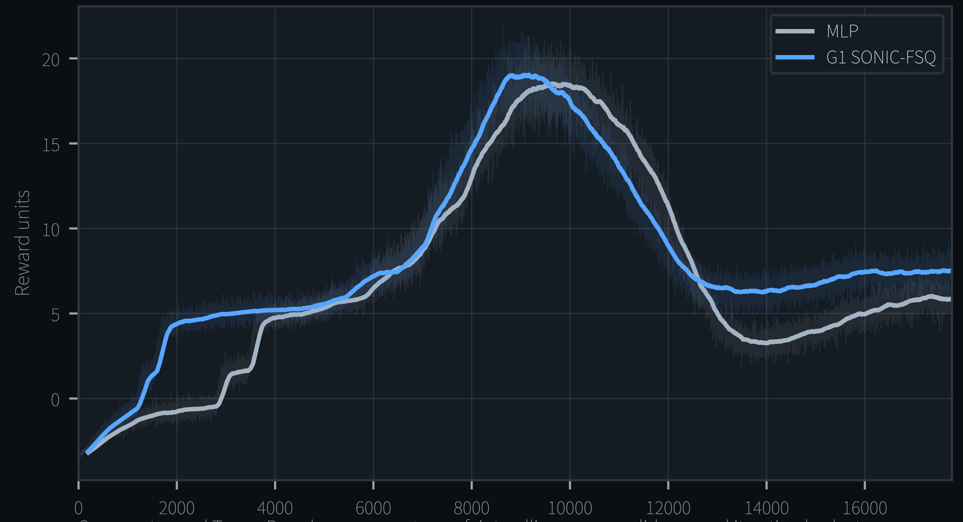
MLP vs G1 SONIC-FSQ — Training Collapse Review

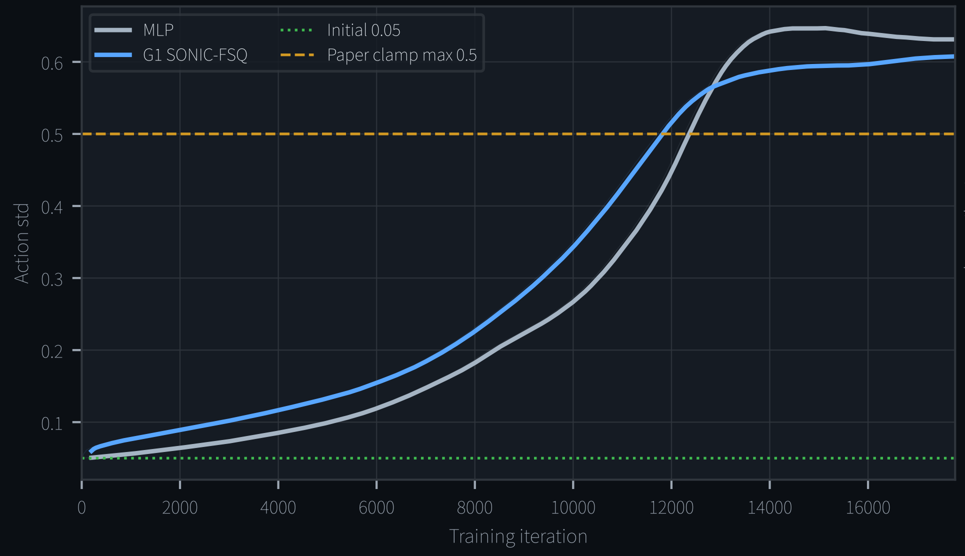
Stopped 23 Jul 2026 • matched comparison through iteration 17,772 • 200-iteration rolling means

Mean episode reward

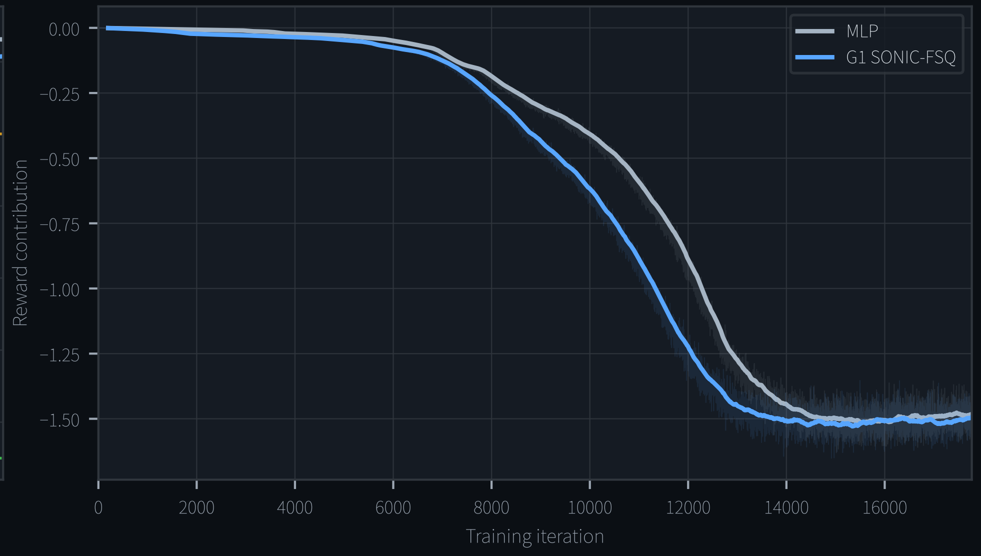


Source: stopped TensorBoard runs • raw traces faint, rolling means solid • equal iteration budget

Mean Gaussian action std



Action-rate reward contribution



Training parameter	Our MLP	Our G1 SONIC-FSQ	Official SONIC
Initial action std	0.05	0.05	0.05
Entropy coefficient	0.005	0.005	0.01 code / 0.013 paper
Std clamp	None	None	[0.001, 0.5] paper
Parallel environments	4,096 total	4,096 total	4,096/GPU × 128 paper
Rollout steps / env	24	24	32 code
Training iterations	100,000 target	100,000 target	100,000 code / 50,000 paper
PPO epochs / mini-batches	2 / 8	2 / 8	5 / 4 code
Learning rate	3e-5 shared	3e-5 shared	actor 2e-5 / critic 1e-3
Clip / γ / λ	0.2 / 0.99 / 0.95	0.2 / 0.99 / 0.95	0.2 / 0.99 / 0.95
Desired KL / max grad norm	0.01 / 1.0	0.01 / 1.0	0.01 / 1.0
Optimizer β	(0.9, 0.98)	(0.9, 0.98)	Not stated in cited config
Actor	MLP 1024–512–256	G1 SONIC-FSQ	Full SONIC actor
FSQ token geometry	N/A	2 × 32-D; 32 levels	2 × 32-D; 32 levels
Auxiliary losses	No	No	Recon + align + cycle
Seed	42	42	Not fixed in cited config

Stopped-run result

MLP

Peak reward 18.52 @ 9,627 | recent 5.88

std 0.631 | episode 365 | action-rate -1.48

G1 SONIC-FSQ

Peak reward 19.03 @ 9,165 | recent 7.50

std 0.607 | episode 376 | action-rate -1.50

Assessment

Both learned and peaked near 9k, then lost about two-thirds of peak reward. Episode length stayed high: this is not a dead policy. The shared failure is std inflation (0.05 to about 0.6), above the paper's 0.5 ceiling, while the action-rate contribution fell to about -1.5.

Recommended next controlled test

Add the paper std clamp [0.001, 0.5]; keep all other settings fixed.

Official sources: released ppo_im_phc.yaml (code values); SONIC paper Table S2 (paper-only values). Our values: saved run config + launch arguments.

Snapshot 2026-07-23 12:29 CST